Phần mềm (Tiết kiệm): Trong CubeMX, tại phần cấu hình GPIO cho PA4 và PB12, bạn chọn Pull-up

MCU

U4 — STM32F103RCT6

XTAL1 HC49S 8MHZ; OSC_IN, OSC_OUT; C17 20pF, C18 20pF; GND

VDD_1, VDD_4, VDD_3, VDD_2, VDDA, VBAT, BOOT0 — +3V3

PA0-WKUP, PA1, PA2, PA3, PA4, PA5, PA6, PA7, PA8, PA9, PA10, PA11, PA12, PA13, PA14, PA15

PB0, PB2, PB1, PB3, PB4, PB5, PB6, PB7, PB8, PB9, PB10, PB11, PB12, PB13, PB14, PB15

PC0, PC1, PC2, PC3, PC4, PC5, PC6, PC7, PC8, PC9, PC10, PC11, PC12

PC13-TAMPER-RTC, PC14-OSC32_IN, PC15-OSC32_OUT

NRST, PD0-OSC_IN, PD1_OSC_OUT, PD2

VSS_2, VSSA, VSS_4, VSS_1, VSS_3 — GND

BOOT0, PA0, RFID_RST, USART2_TX, USART2_RX, RFID_SDA, RFID_SCK, RFID_MISO, RFID_MOSI, TFT_DC, USART1_TX, USART1_RX, TFT_RST, ROW3, SWDIO, SWCLK, ROW2, PB0, BOOT1, PB1, PSM, PB4, PB5, SCL, SDA, PB8, PB9, USART3_TX, USART3_RX, TFT_CS, TFT_SCK, ROW1, TFT_MOSI, NRST, OSC_IN, OSC_OUT, Z-C, PC0, PC1, PC2, PC3, DIM, PC5, PC6, PC7, PC8, ROW4, COL3, PC11, COL2, COL1, PC14, PC15

RESET - BOOT

+3V3; R8 10k; R9 220; NRST; S1; C20 100nF; GND

BOOT0 — Jumper — R14 10k — GND

BOOT1 — Jumper — R15 10k — GND

R16 4.7k, R17 4.7k

LOAD PROGRAM

JP 1x5 2.54: SWCLK, SWDIO, NRST; R22 220; +3V3; GND

FILTER

C28 100nF, C29 100nF, C30 100nF, C31 100nF, C32 100nF, C33 100nF, C34 100nF, C35 100nF, C36 100nF, C37 100nF, C38 100nF, C39 4.7uF — +3V3, GND

FREE

JP 1x6 2.54: PC15, PC14, PA0; +5V, +3V3; GND

NO-PULL_INT

MODULE SIM

STM32 NO-- PULL UP _RX

U3 SIMA7682S(module): VCC, GND, TX, RX, PEN, NET, DTR

USART1_RX, USART1_TX; C15 100nF; +12V; GND

Module sim

TFT

PULL UP _CS; C16 10uF; +5V; +3V3; GND

TFT_SCK, TFT_MOSI, TFT_DC, TFT_RST, TFT_CS

LCD I2C

C19 100nF; SCL, SDA; +5V; GND

ESP32

U5 — ESP32-DEVKITC-32D

3V3, EN, SENSOR_VP, SENSOR_VN, IO34, IO35, IO32, IO33, IO25, IO26, IO27, IO14, IO12, GND1, IO13, SD2, SD3, CMD, EXT_5V

GND3, IO23, IO22, TXD0, RXD0, IO21, GND2, IO19, IO18, IO5, IO17, IO16, IO4, IO0, IO2, IO15, SD1, SD0, CLK

IO32, IO33, IO27, IO23, IO18, USART2_RX, USART2_TX; R12 1k, R13 1k; GND; +5V; C21 10uF, C22 100nF; D4

BUTTON

PUMP: +3V3, R10 10k, PC3, R11 220, C23 100nF, GND

BUZZER: +3V3, R18 10k, PC2, R19 220, C24 100nF, GND

FAN: +3V3, R20 10k, PC1, R21 220, C25 100nF, GND

LIGHT: +3V3, R26 10k, PC0, R27 220, C27 100nF, GND

DIM: +3V3, GND

RFID

RFID_SDA, RFID_SCK, RFID_MOSI, RFID_MISO, RFID_RST; +3V3; C26 100nF; GND

Header 8P 2.54 Duc Thang

KEYPAD 3X4

+3V3; R23 10k, R24 10k, R25 10k

COL1, COL2, COL3, ROW1, ROW2, ROW3, ROW4

Header 7P 2.54 Duc Thang

PZEM-004T

STM32 NO-- PULL UP _RX

+5V; USART3_TX, USART3_RX; 1k R28; C40 100nF; GND

Header 4P 2.54 Duc Thang

FT = Five-volt tolerant. In order to sustain a voltage higher than VDD+0.3 the internal pull-up/pull-dow

I2C HEADER

+5V; SDA, SCL; GND

1-WIRE

PC8 — DHT11 — +3V3, GND

ADC

PB0 — FLAME SENSOR — +3V3, GND

DO

PC7 — HC-SR501 — +5V, GND

DO

PC6 — HC-SR501 — +5V, GND

C41 10uF, C42 100nF; +5V; GND

PC5 — MQ-2; R31 10k; GND, GND

NO-PULL_INT — MẶC ĐỊNH CẤU HÌNH ADC LÀ TẮT PULL-UP/DOWN — PULL_DOWN_INT — PULL_DOWN_INT — NO-PULL_INT — NO-PULL_INT

Altium Designer	Title: **SMART_HOME**	Desc: **MCU.SchDoc**
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	Approved by: *	Sheet **4** of **5**